

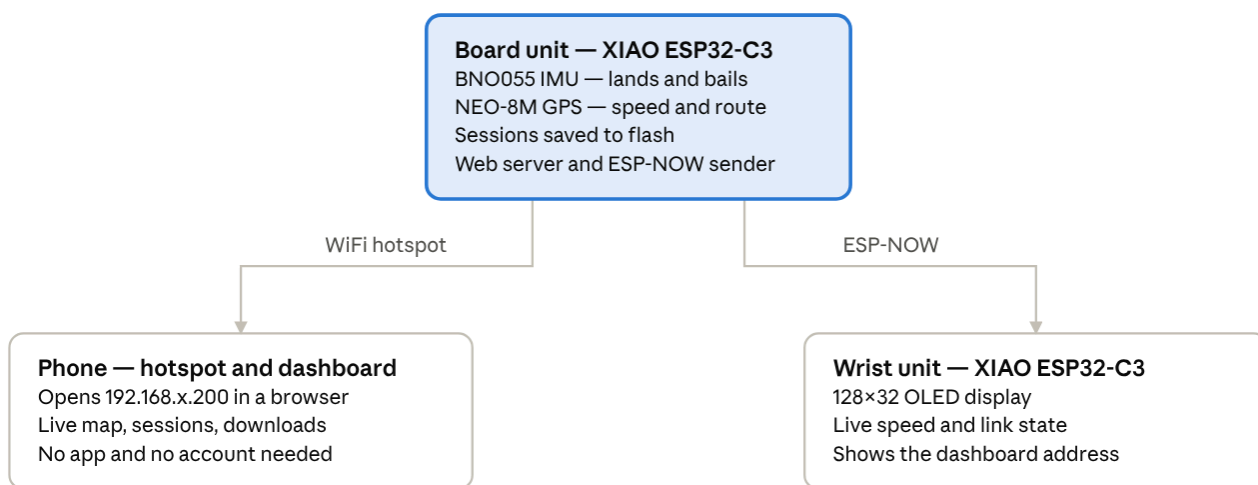
SkateGuard — 6-Week Project Report

Oct 2, 2026 · @Someone

SkateGuard counts kickflip lands and bails, tracks speed, distance and route, and serves its own dashboard from the board to the rider's phone over a hotspot, with a wrist unit showing live speed and the dashboard address. Weeks 1 to 5 were delivered in full and Week 6 is under way, with four design changes made during the build: a BNO055 IMU in place of the MPU6050, ESP-NOW plus a self-hosted web dashboard in place of BLE and a companion app, oscillation-based wobble detection in place of a 25 km/h speed trigger, and bails judged by how the deck settles rather than by an impact spike. This report follows the plan week by week.

At a glance

The board does the work; the phone and the wrist only show it



system overview · board, phone, wrist

The board is the only part that senses, decides and stores; the phone and the wrist are displays. Nothing leaves the rider's hotspot, so the system works with no mobile data and no account.

The six weeks at a glance

Every week's objective was met, several by a different route than planned. Changes from the original plan, near the end of this report, sets out each change and why it was made.

Week	Planned	Delivered
1	Finalise layout, lock the wobble threshold at 25 km/h, confirm the Kickflip approach, procure ESP32, MPU6050, GPS, BLE wearable parts, motor, battery	Layout fixed: board unit under the deck, wrist unit on the strap. Parts bought around two XIAO ESP32-C3 boards, a BNO055 IMU and a NEO-8M GPS. Wobble redefined as an oscillation pattern rather than a speed trigger
2	Mechanical layout, 3D-printed board housing and wrist casing, ESP32 toolchain, basic IMU and GPS communication	Both enclosures built. Arduino IDE with ESP32 core 3.2.0 on XIAO_ESP32C3, IMU reading over I2C and GPS over UART1 at 9600 baud
3	Firmware reading live IMU and GPS, wobble algorithm with the 25 km/h gate, Kickflip detection from rotational signatures	Main loop reading both sensors continuously. Kickflip detection built as a four-stage state machine on the gyro's Y axis. Wobble detection built on counted oscillations, with no speed gate
4	BLE between board and wearable, haptic feedback on wobble, bail detection from impact spikes, land versus bail classification	ESP-NOW link between the two boards, with channel hunting. Land and bail separated by how the deck settles after the rotation, not by an impact spike
5	Companion app or dashboard, GPS route mapping, ride statistics, post-ride summary, live sync over BLE	Dashboard served by the board itself over the phone hotspot: live readout, route map, session list, per-session CSV and route-map download
6	Full system testing across speeds and surfaces, tune sensitivity, finalise accuracy, final assembly, demonstration and documentation	Firmware, dashboard and both enclosures assembled and running. Bench and browser testing complete; on-road tuning of detection sensitivity is in progress

Week 1 — Foundation and design

Planned: finalise the prototype layout, lock the wobble threshold, confirm the kickflip approach and procure the hardware.

Delivered: the layout and the detection approach were fixed, and the parts list was bought around two XIAO ESP32-C3 boards.

- Board unit mounted under the deck on a plate that bolts around the truck baseplate; wrist unit on a strap.
- Parts: two Seeed XIAO ESP32-C3 boards, a BNO055 IMU, a NEO-8M GPS, an SSD1306 128×32 OLED, two 1500 mAh lithium cells, two USB-C charging modules, a vibration motor and indicator LEDs.
- Kickflip detection scoped to rotation on one gyro axis, with the landing judged by how the deck settles rather than by an impact spike.
- Wobble redefined as a counted oscillation pattern instead of a 25 km/h speed trigger, for the reasons in Changes from the original plan.

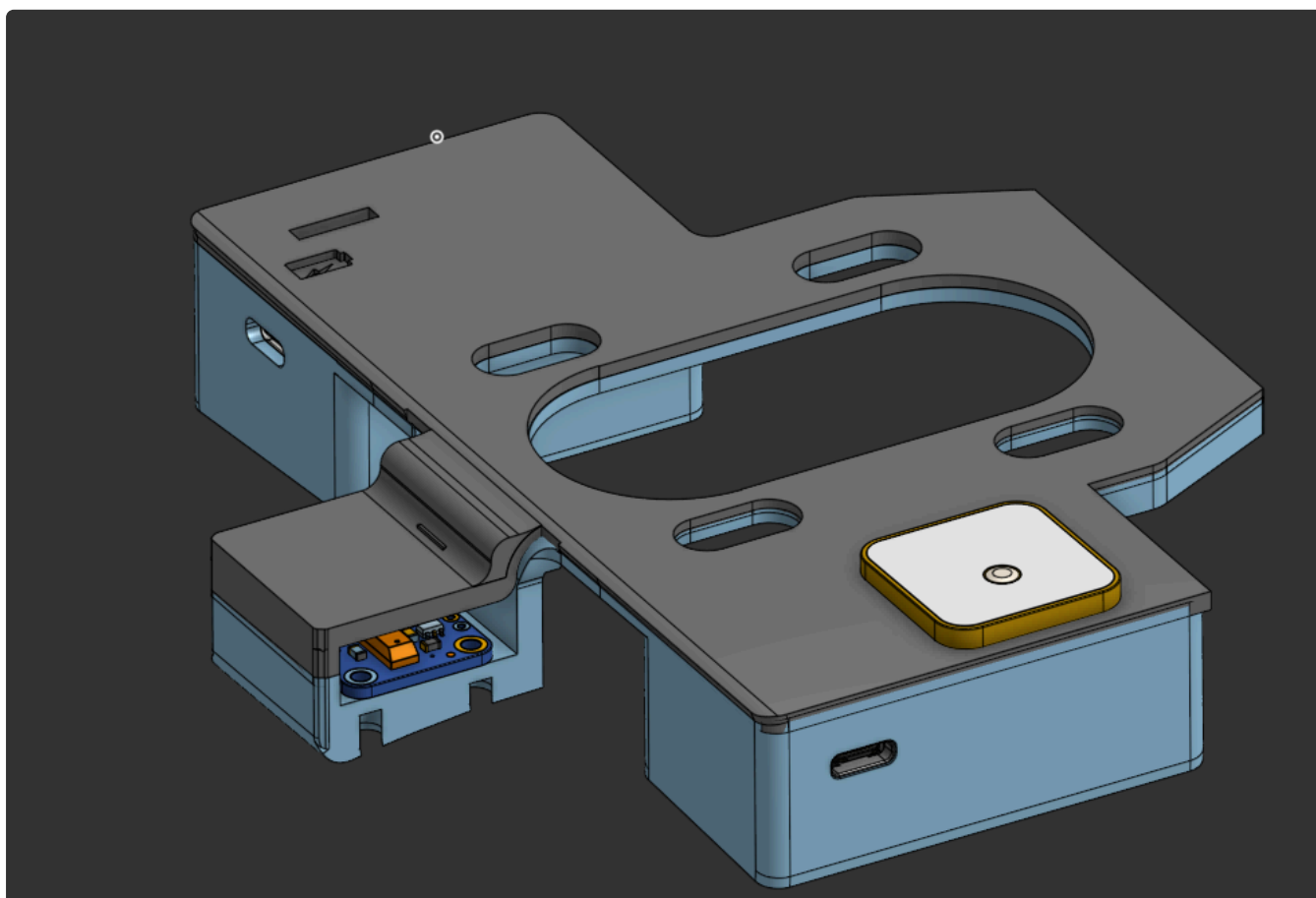
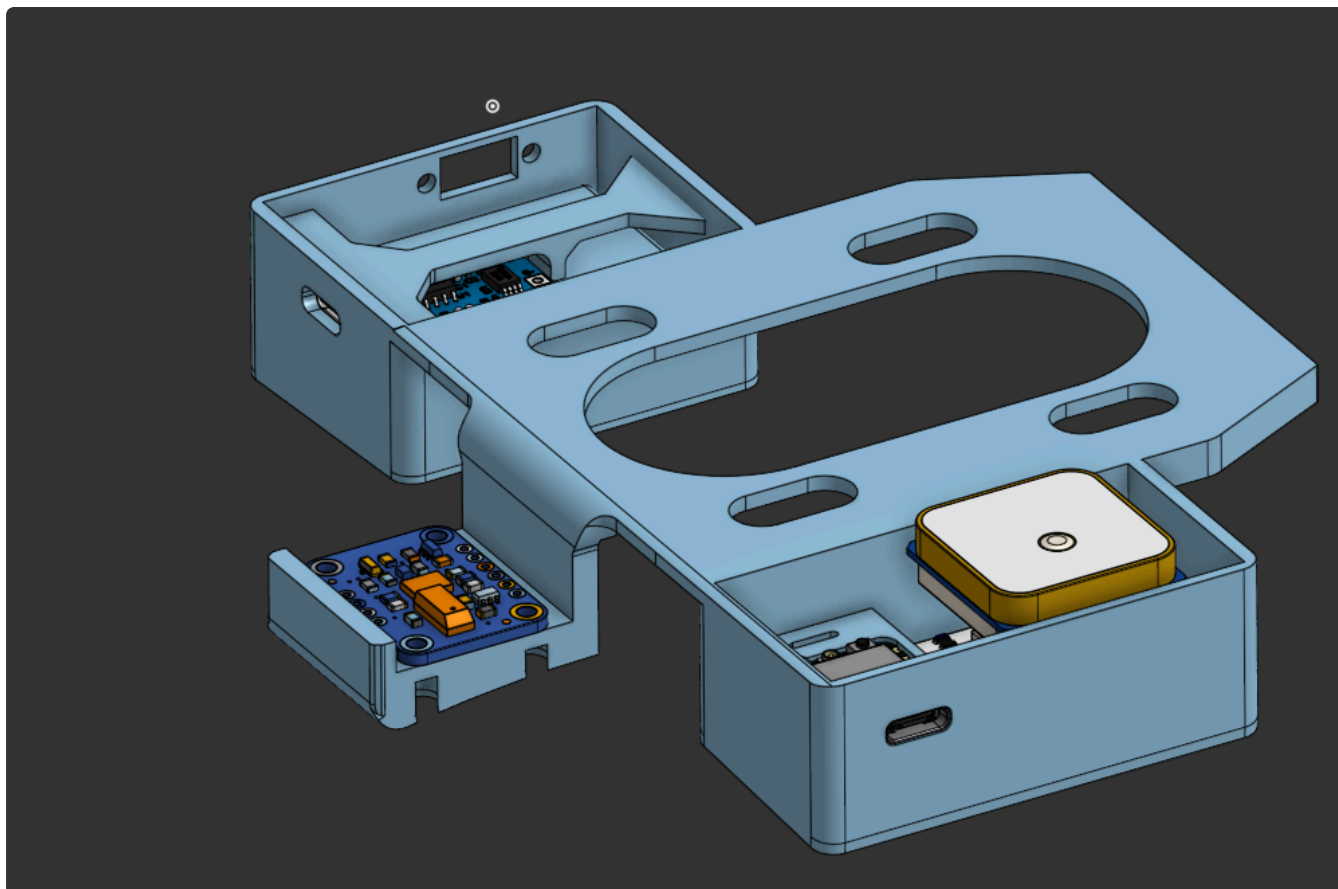
CAD design

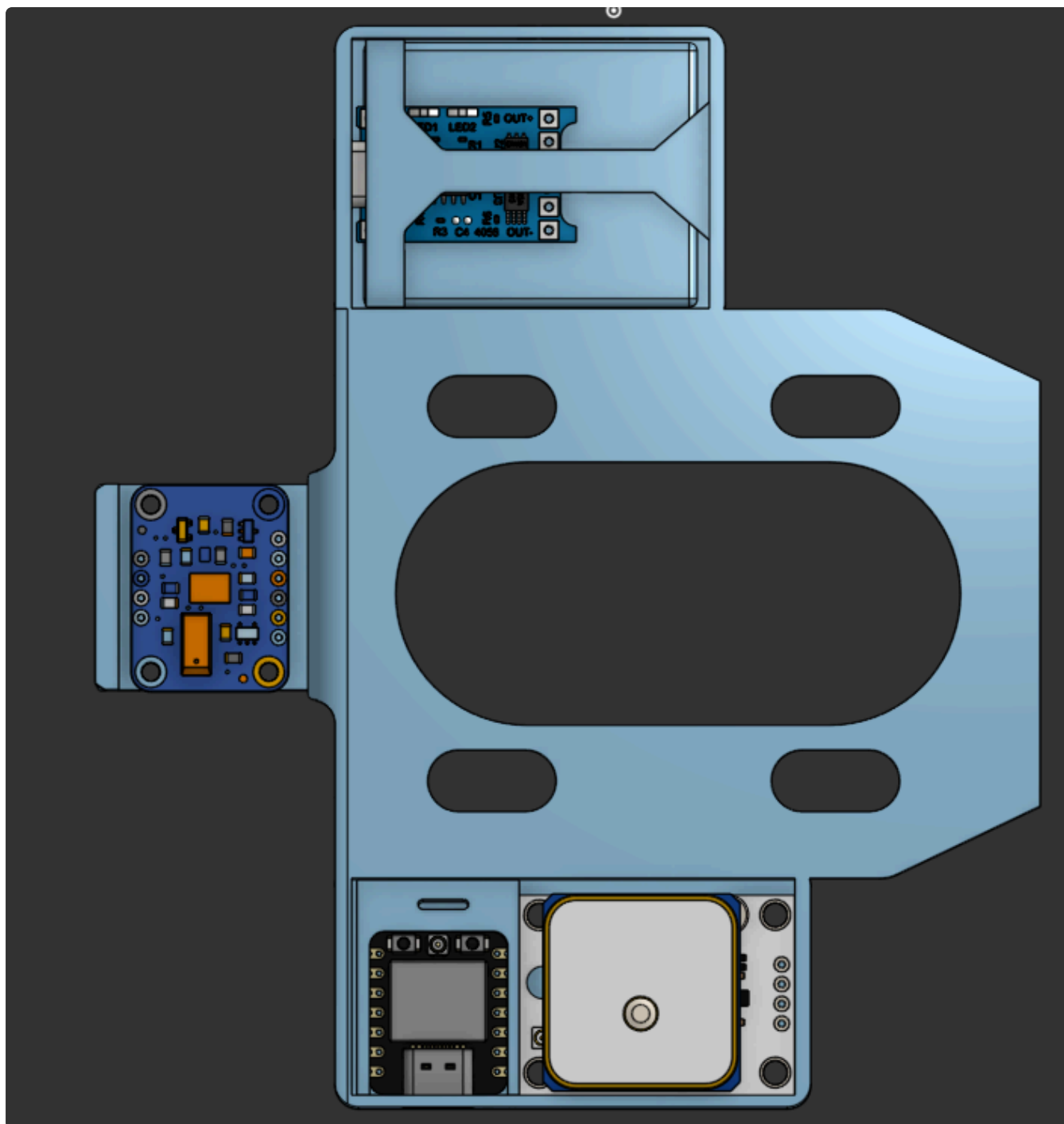
Both enclosures were designed in CAD before anything was printed, so component clearances, port positions and the mounting pattern were settled on screen rather than in filament.

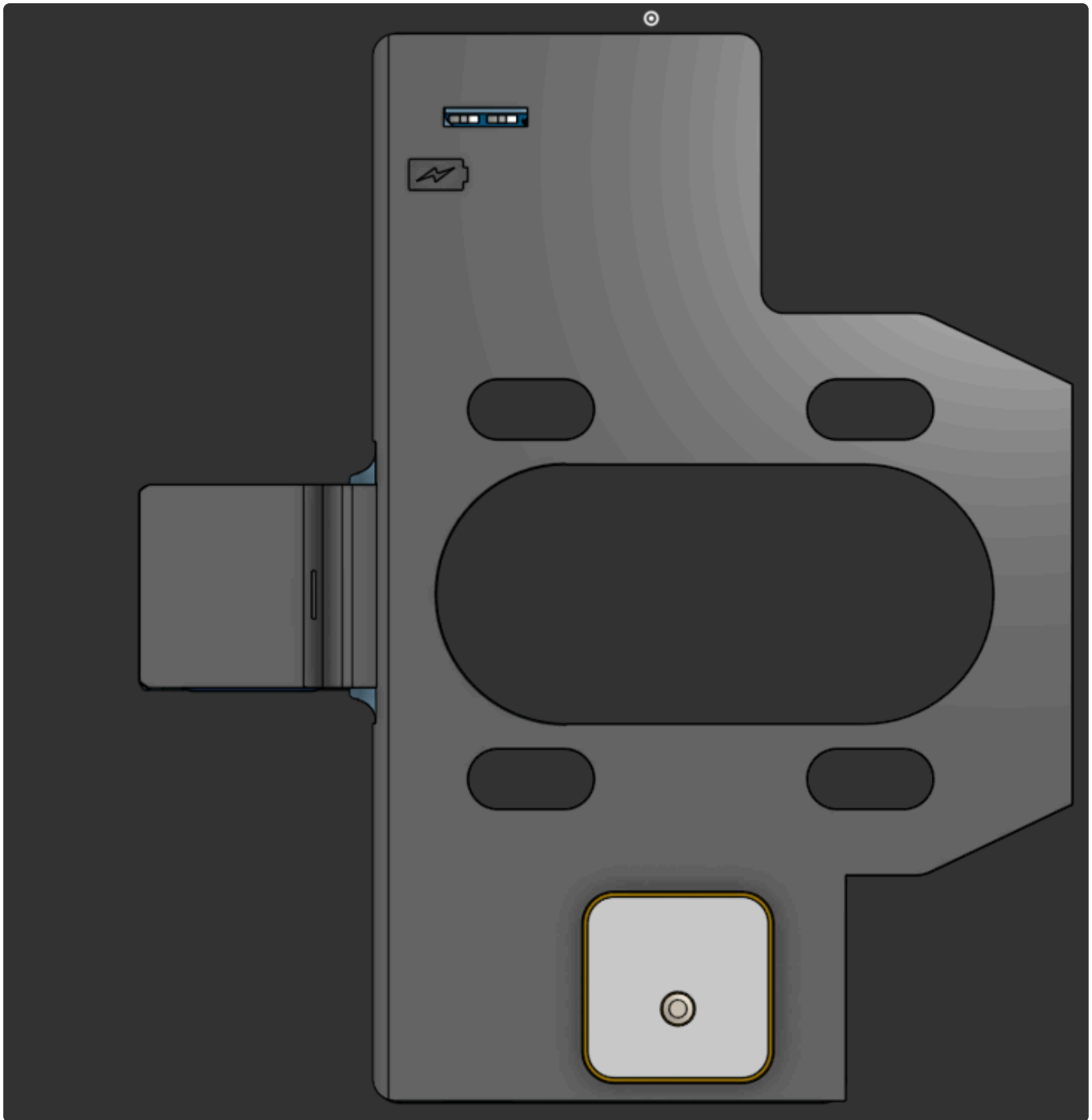
Both units are 3D-printed, two-part enclosures that each hold their board, a lithium cell and a USB-C charging module, with every port and indicator reachable without opening the case.

Board unit

The board enclosure is a mounting plate that sits between the deck and the truck, with the hardware split into three compartments so each part gets what it needs: the GPS patch antenna faces up with a clear view of the sky, the IMU sits in its own pod on the plate's edge where it reads the deck's rotation directly, and the XIAO, cell and charging modules fill the opposite compartment. The oval and slotted cut-outs drop weight and clear the mounting bolts.

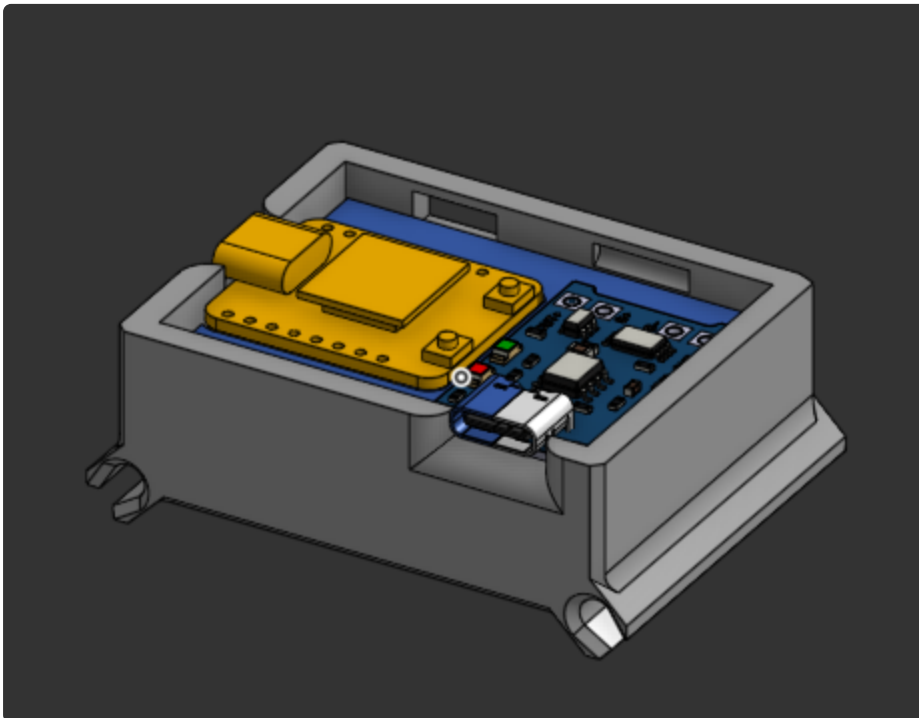
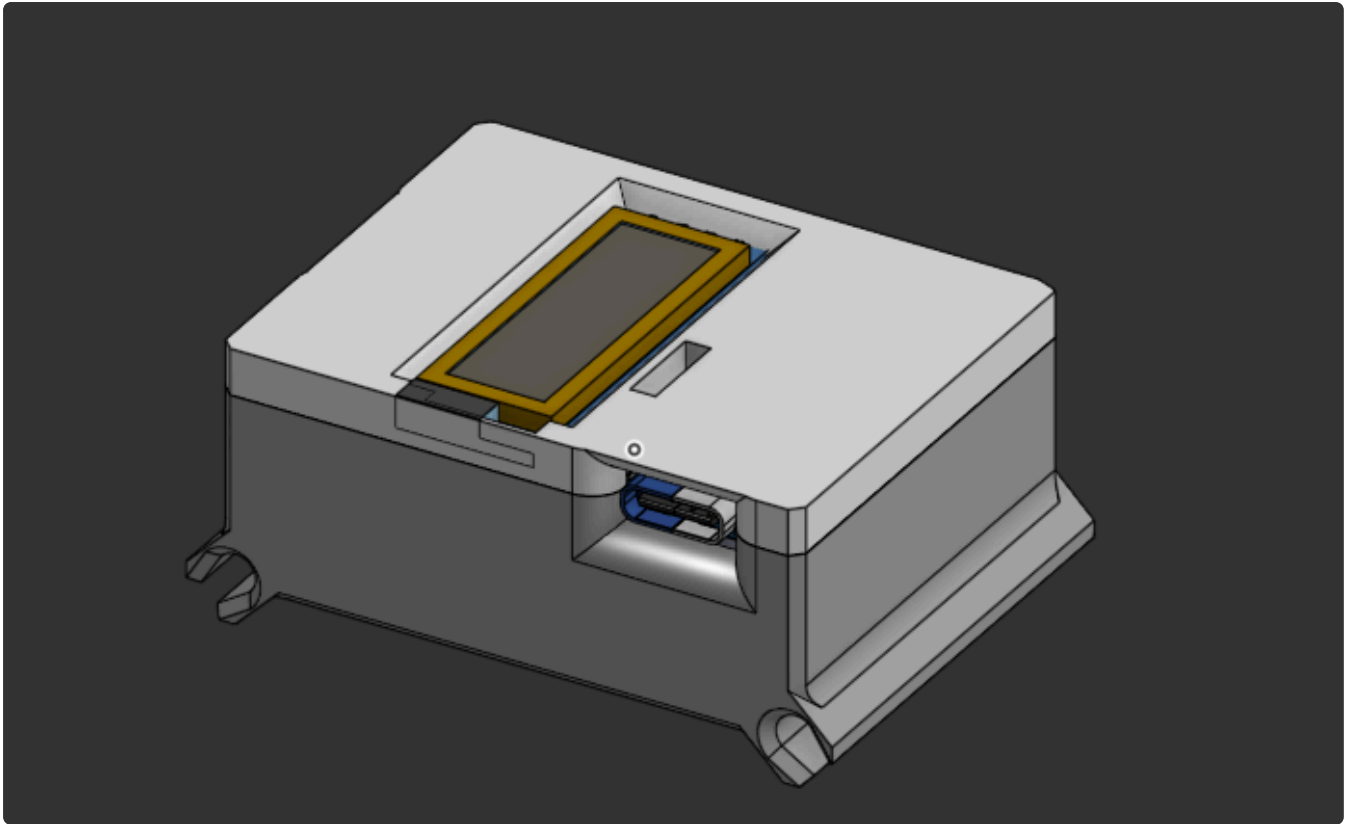


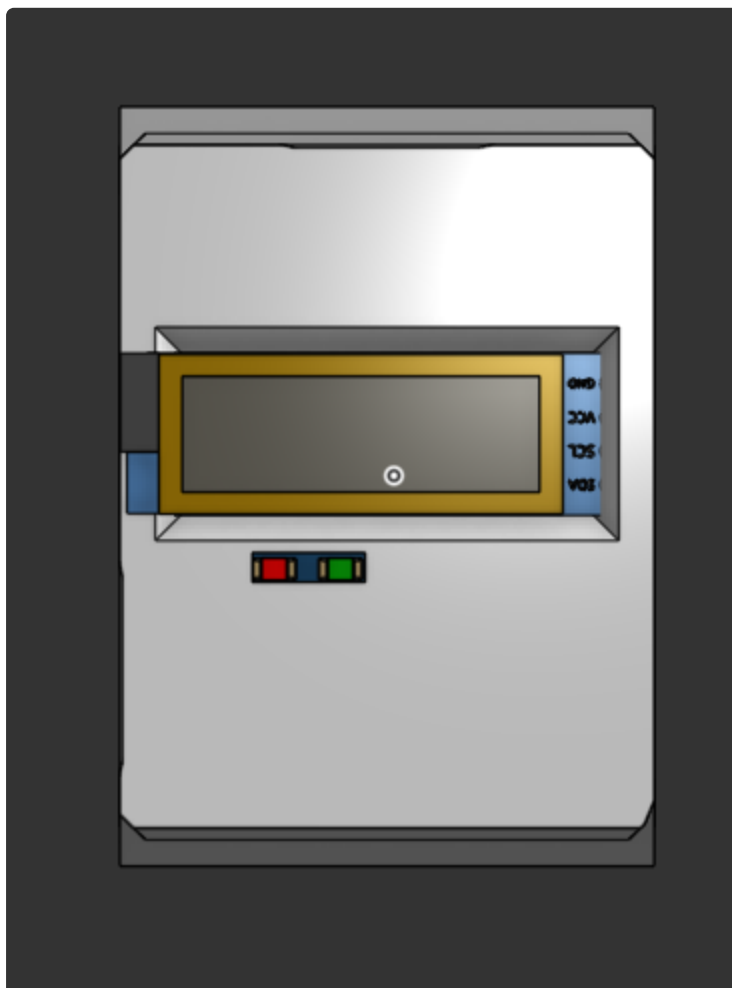
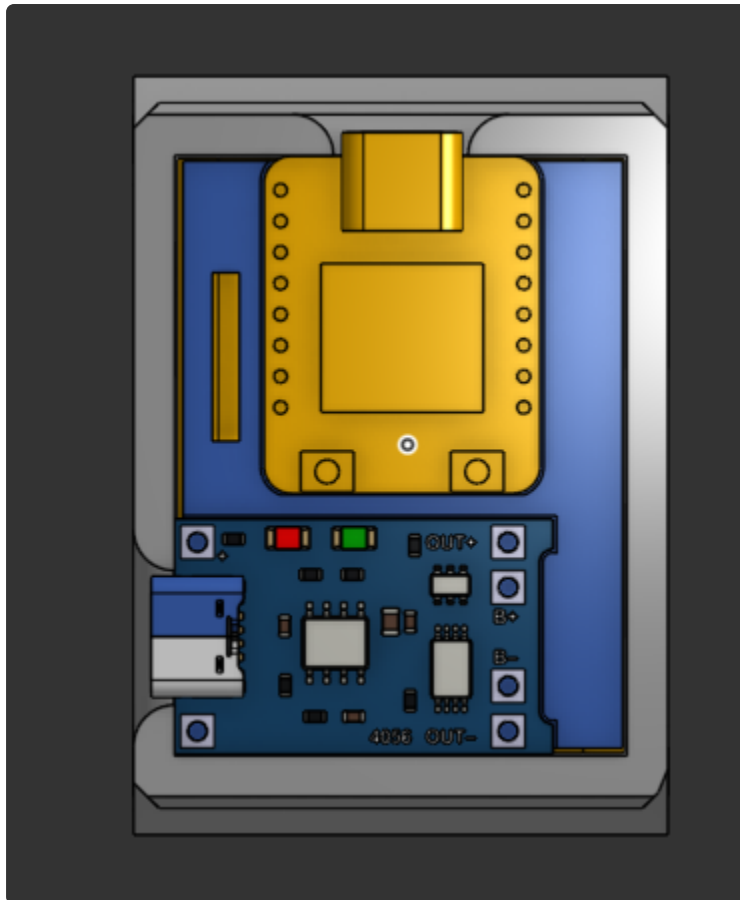


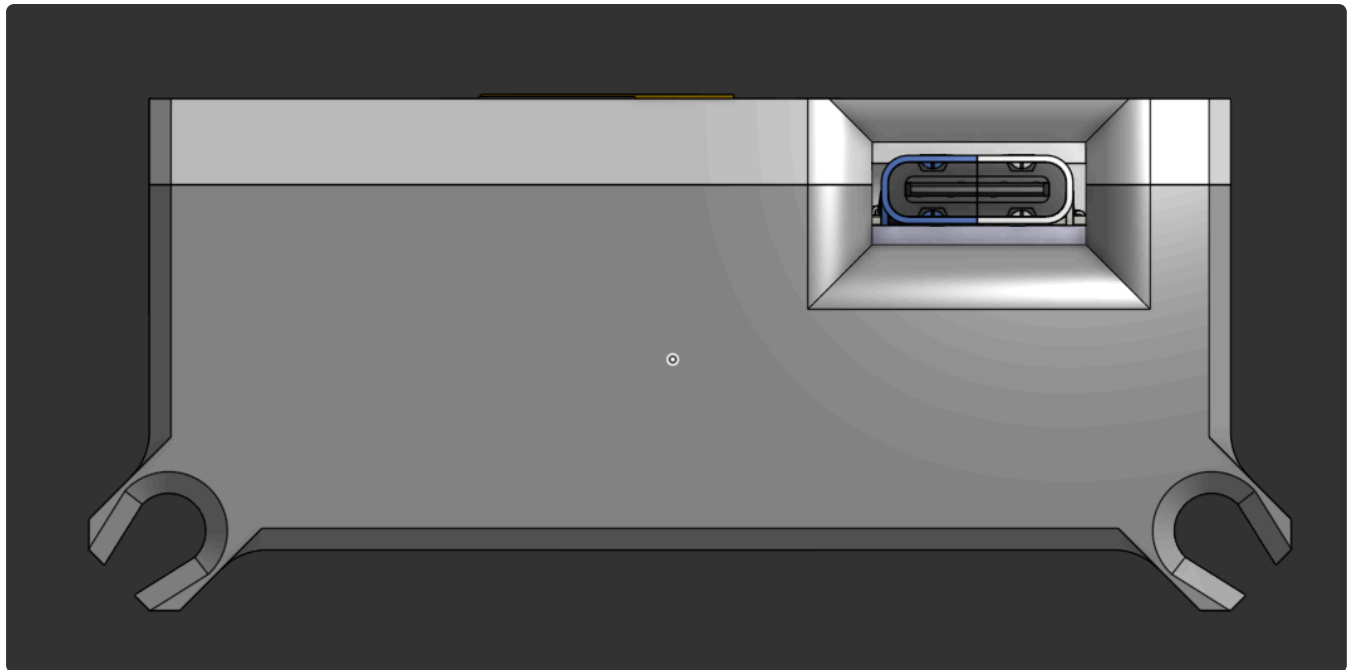


Wrist unit

The wrist enclosure is a lidded box with lugs at all four corners for the strap. The OLED sits in a window in the lid with its four pins (GND, VCC, SCL, SDA) running to the XIAO below, and the two indicator LEDs show through beneath it. The USB-C port of the charging module is recessed into the side wall, so the unit charges on the wrist without being opened.







Week 2 — Hardware and enclosure

Planned: mechanical layout for the IMU and GPS, 3D-printed board housing and wrist casing, ESP32 toolchain, basic IMU and GPS communication.

Delivered: both enclosures printed and populated, the toolchain running on Arduino IDE with ESP32 core 3.2.0 on XIAO_ESP32C3, and both sensors talking to the board.

Both units are Seeed XIAO ESP32-C3 boards, so one toolchain and one radio stack covers the whole system.

Unit	Component	Connection
Board	Seeed XIAO ESP32-C3	Mounted under the deck
Board	BNO055 IMU	I2C — SDA GPIO6 (D4), SCL GPIO7 (D5), address 0x28, fused mode
Board	NEO-8M GPS	UART1 — RX GPIO20, TX GPIO21, 9600 baud
Board	Battery sense	ADC GPIO4 (ADC1_CH4), divider 30k / 10k
Board	1500 mAh lithium cell, USB-C charging module, slide switch	Cell to charger, charger output through the switch to the XIAO
Wrist	Seeed XIAO ESP32-C3	Worn on the strap

Unit	Component	Connection
Wrist	SSD1306 128×32 OLED	I2C — SDA GPIO6 (D4), SCL GPIO7 (D5), driven by U8g2
Wrist	Vibration motor	GPIO3
Wrist	Status LED	GPIO8
Wrist	Battery sense	ADC GPIO2 (D0), divider R1 = 30k / R2 = 10k, 11 dB attenuation
Wrist	1500 mAh lithium cell, HW-373 V1.2 USB-C charging module, slide switch	Same arrangement as the board unit
Link	ESP-NOW broadcast	Channel follows the phone hotspot

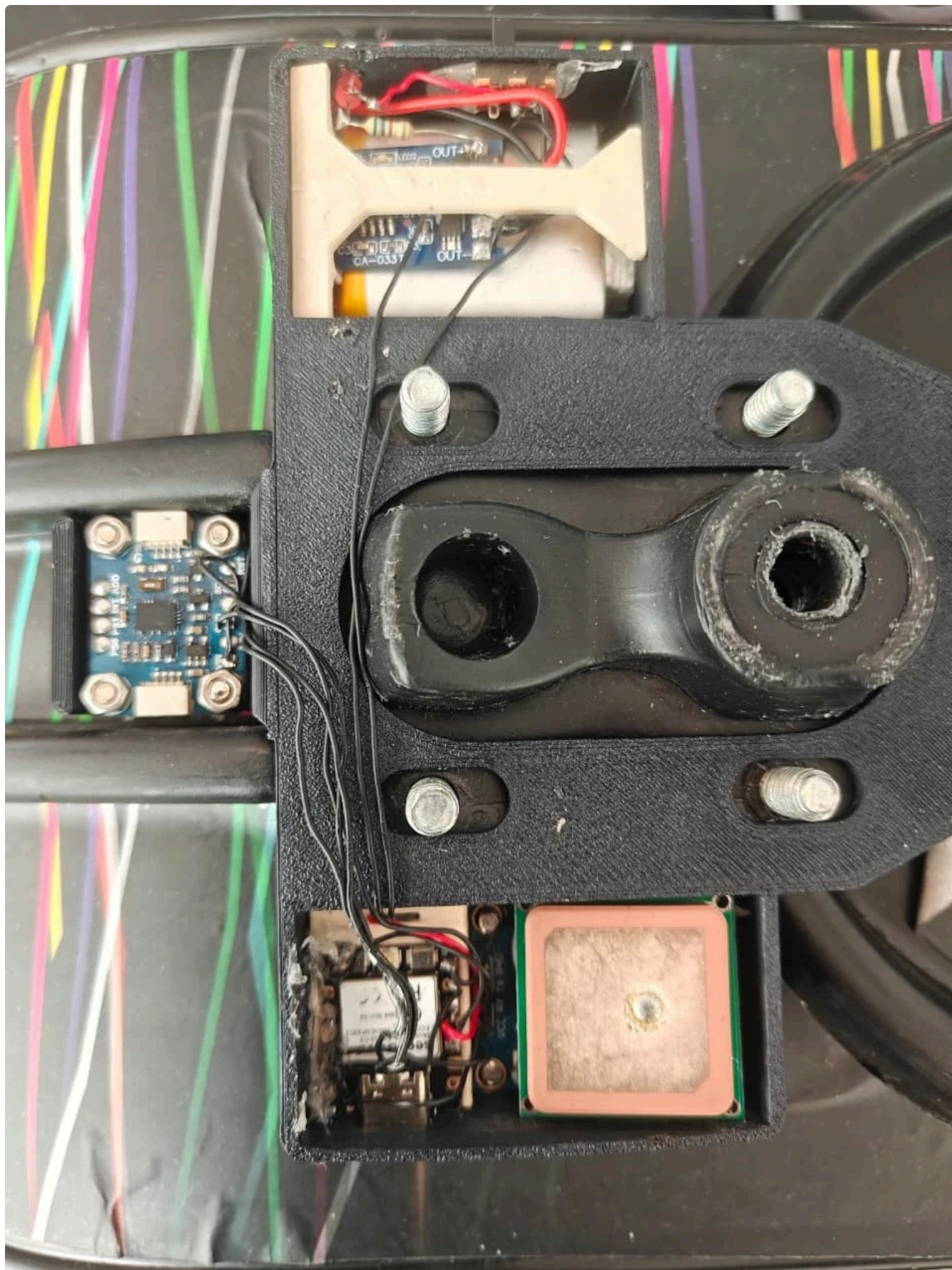
The BNO055 runs its own sensor fusion, so the ESP32 reads finished Euler angles and a gyroscope vector instead of filtering raw accelerometer data itself. Battery voltage is averaged over several ADC reads, because a single read on this chip swings the reported percentage by several points.

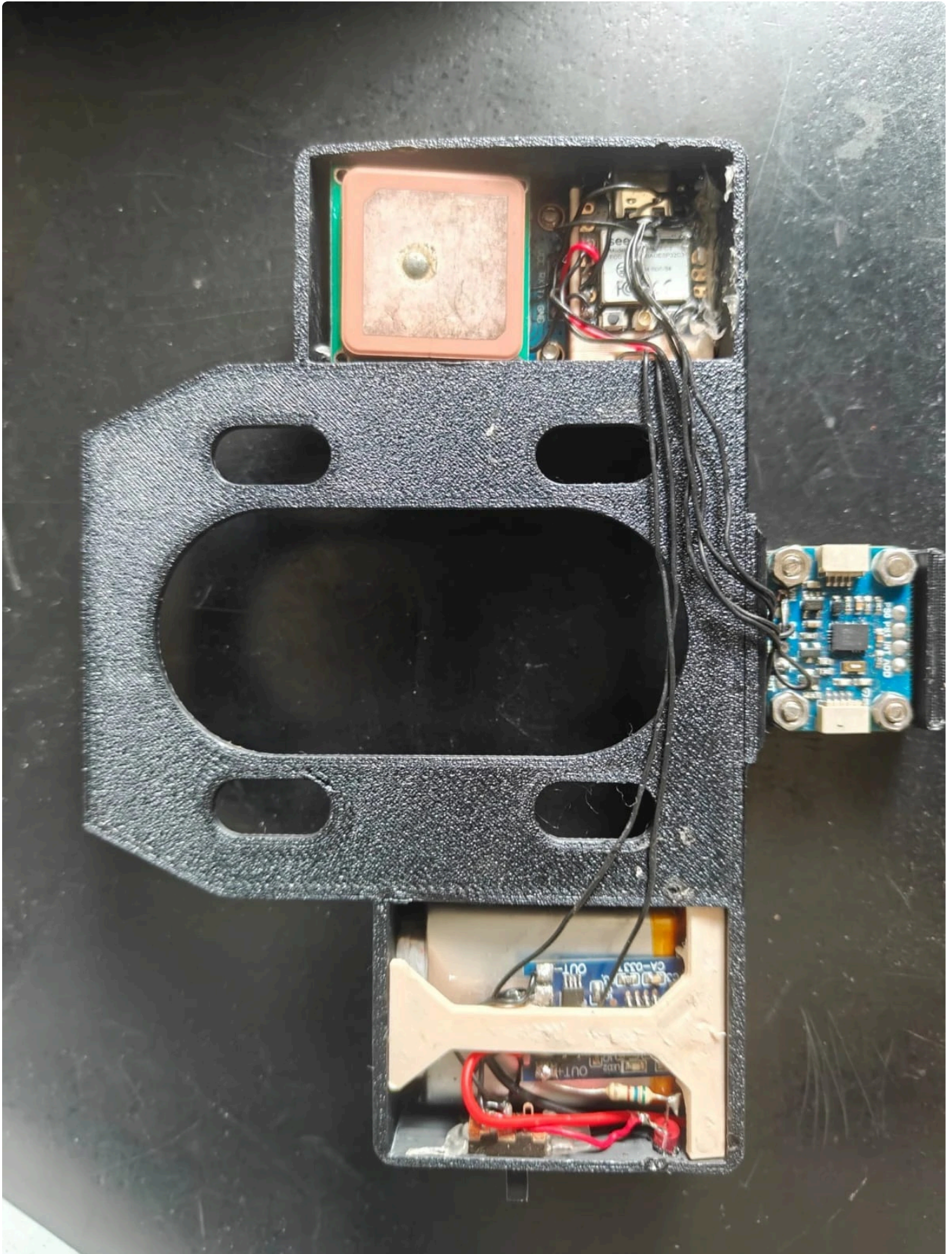
The board sketch uses 91% of the default 1.25 MB program partition and 12% of RAM. Session data lives in a separate LittleFS partition, so a nearly full sketch cannot push the stored rides out.

Printed, populated and wired

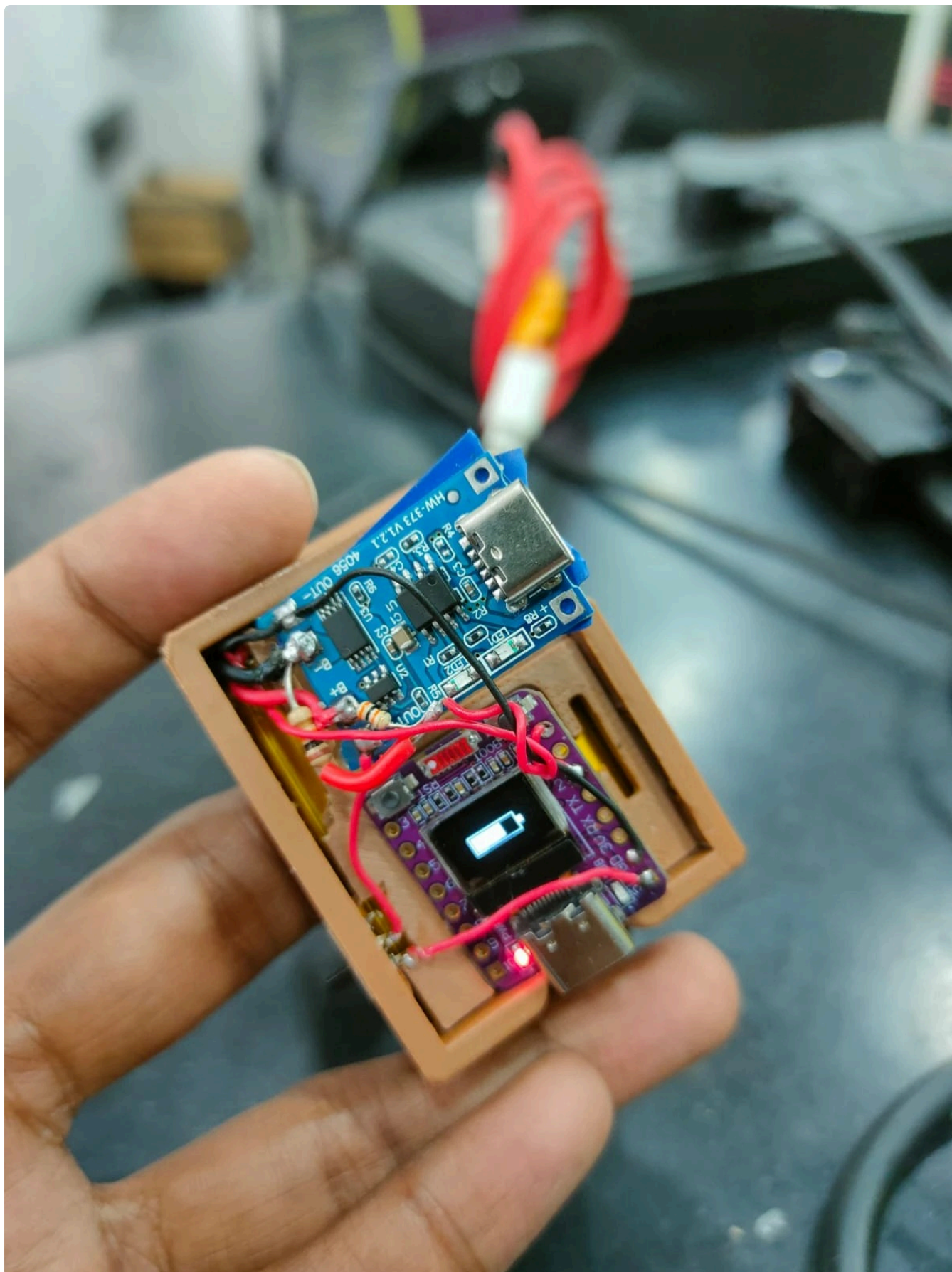
The CAD in Week 1 came out as two printed enclosures, populated and mounted.

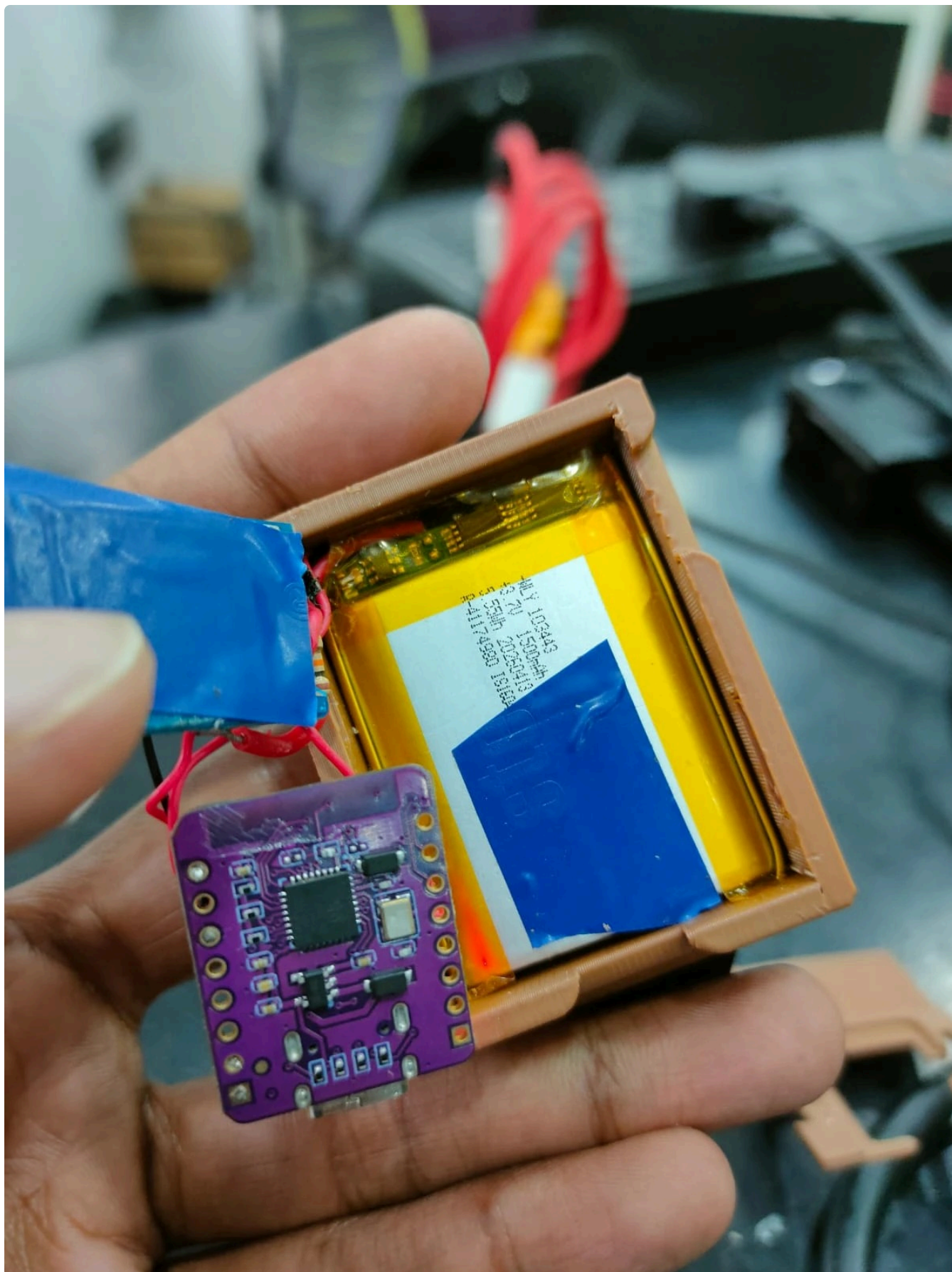
Built and mounted, the plate bolts around the truck baseplate with the IMU pod hanging off the deck's edge. The charging module, cell and power switch sit in the upper compartment, the XIAO and the GPS antenna in the lower one.





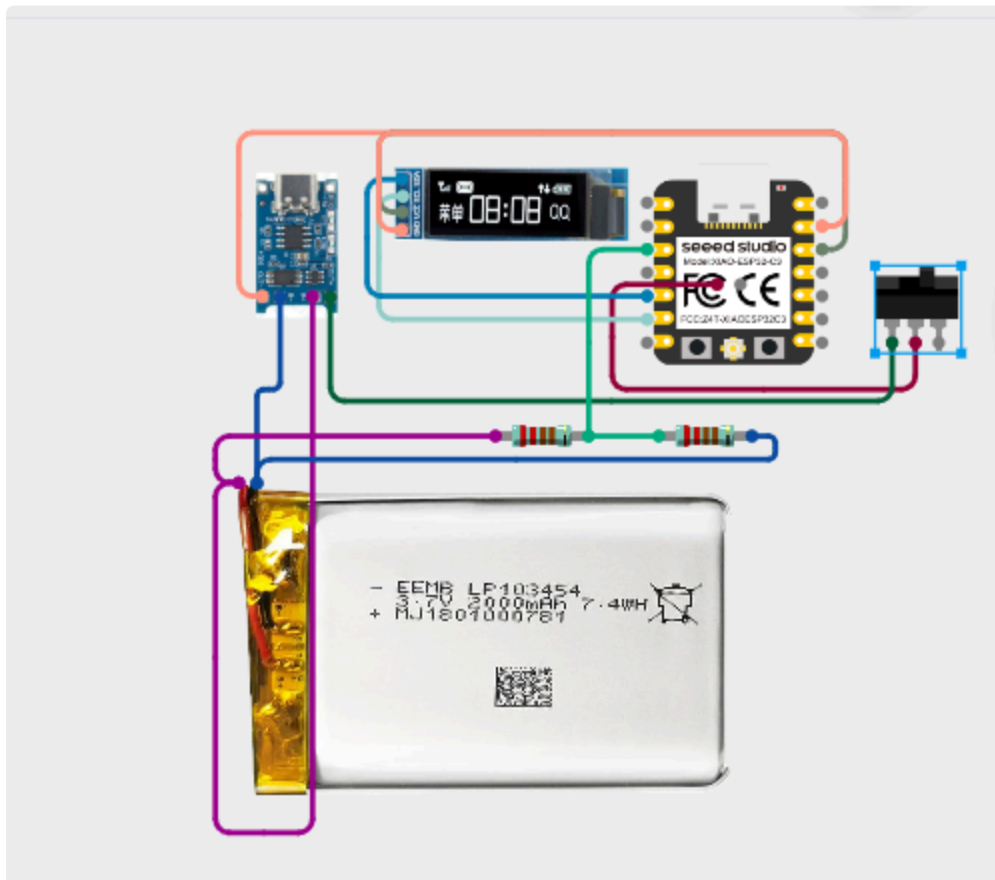
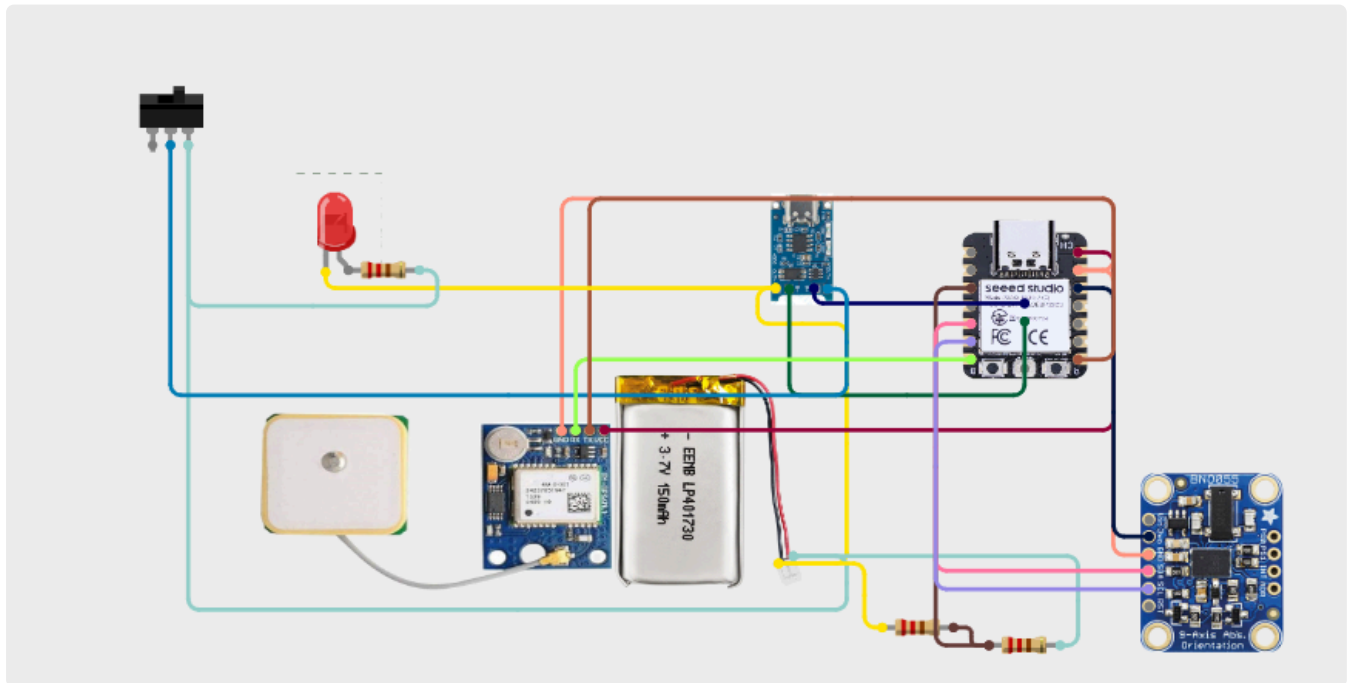
The built unit uses an HW-373 V1.2 USB-C charging module (TP4056 class) stacked over the XIAO, with the OLED facing out through the lid.





Wiring

Each unit is wired the same way: the cell feeds a USB-C charging module, the module's output passes through a slide switch to the XIAO, and the sensors or display hang off the XIAO's I2C and UART pins. The resistors in each diagram are the I2C pull-ups and the indicator-LED series resistor. Both units run a 3.7 V 1500 mAh lithium cell; the capacities printed on the diagram artwork are stand-ins from the drawing tool.



Signal	Board unit	Wrist unit
I2C data	GPIO6 (D4) to BNO055	GPIO6 (D4) to OLED
I2C clock	GPIO7 (D5) to BNO055	GPIO7 (D5) to OLED
Serial	GPIO20 RX, GPIO21 TX to NEO-8M at 9600 baud	—
Battery sense	GPIO4 (ADC1_CH4) through a 30k/10k divider	GPIO2 (D0) through a 30k/10k divider
Motor	—	GPIO3
Status LED	Indicator LED on the charging module	GPIO8

The I2C pins and the board unit's ADC pin are marked **UNVERIFIED** in the firmware, so confirm them against these diagrams before the pin table is treated as final.

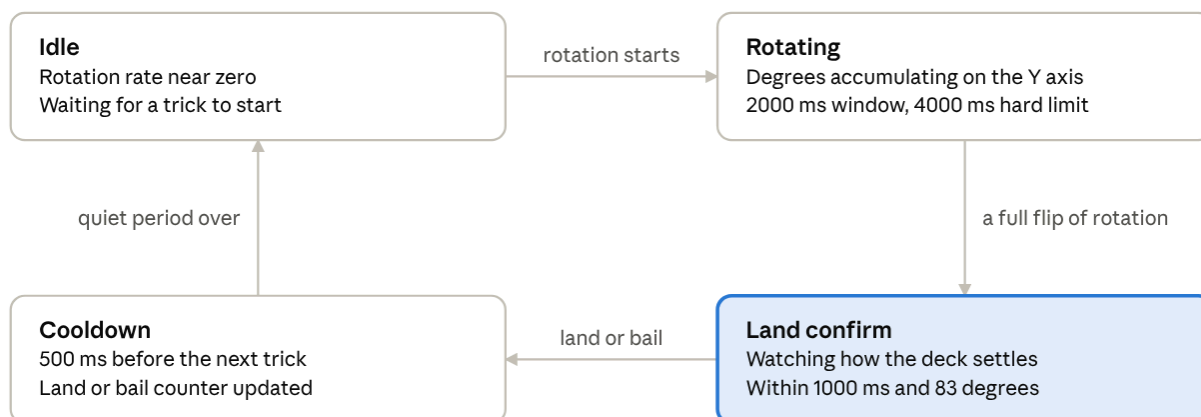
Open question: the print material, layer height, infill and finished dimensions of each enclosure are not recorded yet.

Week 3 — Sensor and embedded system

Planned: firmware reading live IMU and GPS data, the wobble algorithm, and kickflip detection from rotational signatures.

Delivered: a continuous sensor loop, kickflip detection as a four-stage state machine, and wobble detection on counted oscillations.

A flip counts as a land only if the deck settles within 83 degrees



trick detection · four stages

A trick that reaches Land confirm and settles inside the limit counts as a land. One that keeps swinging past 83 degrees, or runs out of time, counts as a bail. Speed wobble is handled separately: it needs three counted oscillations before it is reported, so a single swerve does not trigger an alert.

Constant	Value	Purpose
MOTION_WINDOW_MS	2000	How long the active rotation window stays open
MOTION_ABSOLUTE_TIMEOUT_MS	4000	Hard limit on any single trick
MOTION_COOLDOWN_MS	500	Quiet period after a counted trick
LAND_CONFIRM_WINDOW_MS	1000	Time allowed to confirm the landing
LAND_CONFIRM_MAX_DEG	83.0	Swing past which the landing is rejected as a bail
WOBBLE_OSCILLATIONS_REQUIRED	3	Oscillations before a wobble is called

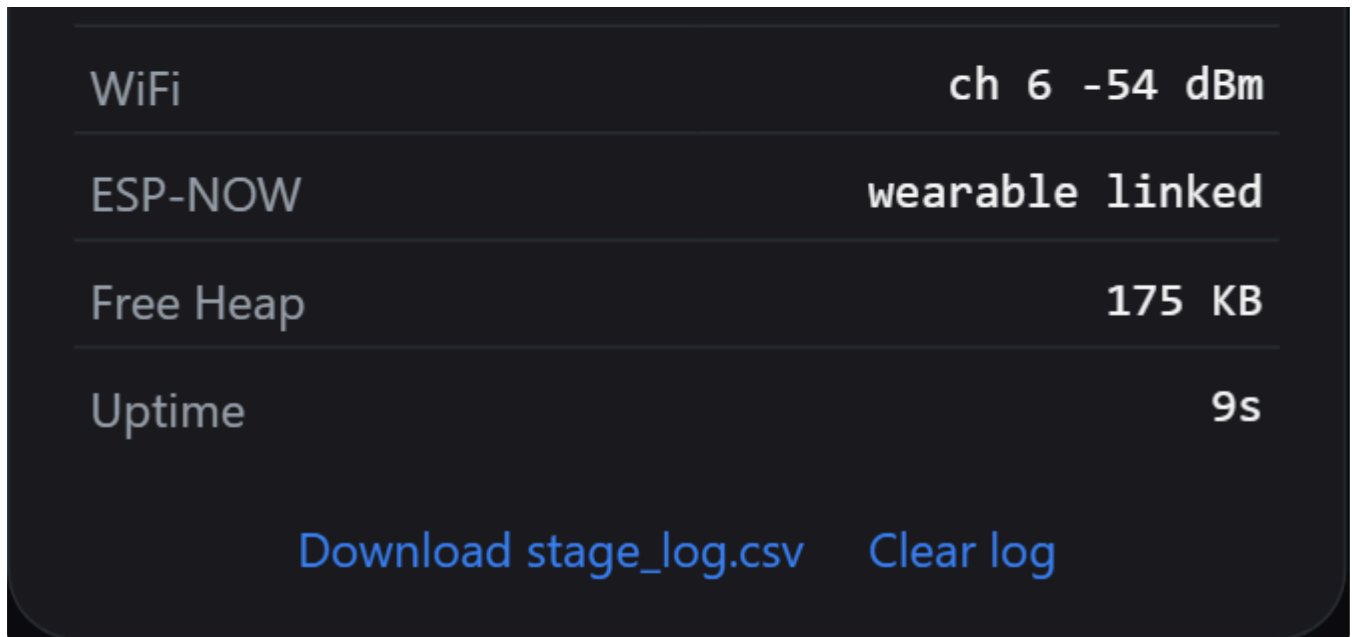
A GPS fix is trusted only when the position is valid, less than 3 seconds old and backed by at least 4 satellites. Speeds under 2 km/h report as zero so a parked board does not drift, and a position step implying more than 20 m/s is rejected as multipath.

The debug page shows the same values live, twice a second, which is what the thresholds were set against.

Debug



Stage	IDLE
Gyro Y	-0.9 deg/s
Accum Rotation	0.0 deg
Land Confirm Swing	0.0 deg
Active Window	0 / 2000 ms
Absolute Window	0 / 4000 ms
Cooldown Window	0 / 500 ms
Oscillations	0 / 3
Loop Max / Avg	12 / 3.8 ms
Land Count	1
Bail Count	0
Battery	86% (4.02 V)
GPS Satellites	9
GPS HDOP	0.9
GPS Fix Age	0.4 s
GPS Data In	9426 chars
GPS Checksum Fails	0



Week 4 — Wireless alerts and bail detection

Planned: a wireless link between board and wearable, haptic feedback on confirmed wobbles, and bail detection separating landings from bails.

Delivered: an ESP-NOW link with channel hunting, the wrist motor and LED pulsing on each wobble the board confirms, and bails separated from lands by how far the deck swings after the rotation.

The board broadcasts telemetry over ESP-NOW, which runs alongside its hotspot connection and needs no pairing. Its peer is registered on channel 0, so it follows whatever channel the phone picks.

The wrist unit does not know that channel at power-up, so it hunts: it parks on each channel for 300 ms until packets arrive, then reads the channel the board states inside the packet and retunes to it. That second step was added after the receiver locked onto a neighbouring channel through adjacent-channel leakage, where packets arrived often enough to look correct and then stopped.

State	Wrist display
Hunting	SCAN ch<n>
Linked, stopped, first 20 s	Dashboard: and the address, with a countdown
Linked, moving	Live speed in large digits
Linked, stopped after the window	Battery icon
Silent for 5 s	NO SIGNAL , then hunting again

The address window is gated on being stopped as well as on the timer, so rolling away inside the window hands the screen straight back to speed.

Image — The wrist unit showing the dashboard address

Serial printing was moved out of the ESP-NOW receive callback into a drained queue. Printing inside the callback blocks the WiFi task long enough to drop the following packets, which made a healthy radio link look unreliable.

Week 5 — Companion dashboard and data logging

Planned: a companion app or dashboard, GPS route mapping, ride statistics, a post-ride summary and live sync.

Delivered: a dashboard served by the board itself over the phone hotspot, with a live map, session recording, a post-ride summary covering distance, top speed, wobble alerts and trick counts, and per-session downloads.

The board joins the rider's phone hotspot, takes a DHCP lease, then moves itself to `<subnet>.200` and verifies the move, so the dashboard address is predictable on any phone. It also answers to `skateguard.local`, though phone support for that is inconsistent, which is why the wrist unit displays the numeric address.

The page is stored in flash inside the sketch and served whole; the browser polls `/api/live` twice a second.

Endpoint	Purpose
<code>/</code>	The dashboard page
<code>/api/live</code>	Telemetry JSON, polled twice a second
<code>/api/debug</code>	Motion state machine, GPS, radio and memory values
<code>/api/sessions</code>	Saved sessions and their routes
<code>/api/command?cmd=</code>	<code>start</code> , <code>stop</code> , <code>save</code> , <code>discard</code> , <code>delete:<n></code>
<code>/log</code> , <code>/clear</code>	Download or clear the motion log
<code>/manifest.json</code>	Web app manifest for "Add to Home screen"

Image — The dashboard on the phone, waiting to record

Stopping a ride does not save it: the card offers Save or Discard, and nothing is written until one is chosen. A saved ride stores its distance, average speed, top speed, land count, bail count and wobble-alert count. Saved rides go to LittleFS, the last 7 kept, with a route point every 30 seconds up to 20 points. Each row can be downloaded as a CSV of the ride plus a rendered PNG of its route, drawn in the browser over OpenStreetMap tiles.

Image — Recording in progress

Image — Stopped, offering Save or Discard

Image — The session list after saving

Route line N belongs to session N, with a `-` placeholder for rides recorded without GPS. An earlier version skipped those blank lines, which slid every later route onto the wrong session.

Week 6 — Testing, tuning and demonstration

Planned: full system testing across speeds and surfaces, tuning to cut false triggers, final assembly, demonstration and documentation.

Delivered so far: both units assembled and running, with bench, browser and first field testing complete. On-road tuning and the demonstration are the remaining work.

Testing so far covers the sensors, the radio and the dashboard. Detection accuracy on real rides has not been quantified yet; that is the open item.

Area	Method	Result
Trick detection	<code>stage_log.csv</code> , a row every 500 ms holding stage, all three gyro axes, accumulated degrees, each window against its limit and the event that fired	Thresholds set to the values in the table above; log downloads cleanly and is capped at 2000 rows

Area	Method	Result
Loop health	Slowest and average loop period recorded and shown on the debug page	Instrumented, because a stalled loop shows up as missed tricks rather than as a timing symptom
GPS	Status line every 5 seconds: satellites, HDOP, characters processed, fix age, checksum failures; first fix after boot timed	Fix acquired outdoors with 4 or more satellites; no fix indoors, as expected
Radio link	Power-up in both orders, walking out of range until NO SIGNAL and back, changing the hotspot so the channel moves	Wrist unit re-finds the board without a reboot in each case
Dashboard	Exercised in a browser against a simulated board: save and discard, the 7-session cap, deleting sessions, downloads with and without a route	All paths work; the route-alignment fault was reproduced and fixed with a no-GPS session saved between two GPS sessions
Field use	Real rides recorded, then opened as live map and session route maps	Routes render over OpenStreetMap tiles and match the ride

Image — A recorded session drawn over OpenStreetMap tiles

Image — The live map during a real ride

What remains for Week 6 is riding sessions with a hand-counted tally of flips, so land and bail counts can be scored against what actually happened and the thresholds tuned to cut false triggers.

Changes from the original plan

Four decisions made during the build differ from the Week 1 plan. Each was a change of method, not of objective.

Planned	Built	Reason
MPU6050 IMU	BNO055 IMU	The BNO055 fuses on the sensor and outputs absolute orientation, so rotation is measured directly instead of being filtered on the ESP32 inside the control loop
BLE link to a companion app	ESP-NOW to the wrist unit, dashboard served by the board	A companion app has to be installed and kept in step with the firmware. A page served from the board opens in any phone browser, and ESP-NOW gives a board-to-wrist link with no pairing step
Wobble triggered above 25 km/h	Wobble from three counted oscillations, no speed gate	GPS speed needs 4 satellites and lags real movement, and wobble can start below 25 km/h. The oscillation pattern is the symptom itself, so it is what the detector watches
Bail detected from an accelerometer impact spike	Bail detected from how far the deck swings after the rotation	A clean landing can produce a sharp impact and a bail can produce none, so the spike does not separate them. The settling swing does

The 25 km/h figure is still usable as a reporting threshold rather than a trigger, if the project needs it stated that way, as a configurable alert level rather than a detection gate.

Limitations and next steps

- Detection is tuned for kickflips. Other tricks rotate around different axes and are not classified.
- The dashboard has no login. Anyone on the hotspot can open it, record and delete sessions; the hotspot password is the only barrier.

- A GPS cold start can take several minutes, and nothing is reported until 4 satellites are visible.
- Map tiles come from OpenStreetMap, so the phone needs internet for the map screens. Recording itself does not.
- The sketch fills 91% of the default 1.25 MB partition. Larger features need a different partition scheme, which erases stored sessions.
- Only the last 7 sessions are kept, and a route point every 30 seconds means the drawn line cuts corners on fast laps.
- Deleting a session happens immediately, with no confirmation.

Next steps, in order:

1. Confirm the rotation axis from one real kickflip, comparing the three gyro axes now logged, and confirm the I2C and ADC pins from the boot log.
2. Ride with a hand-counted tally of flips, score land and bail counts against it, and tune the thresholds using `stage_log.csv` to cut false triggers.
3. Confirm the wobble oscillation count on a real speed wobble rather than a bench shake.
4. Decide whether the 25 km/h figure returns as an alert level.
5. Battery-life measurement for both units over a full session.